



SIMATS Online Programming Lab

JAVA PROGRAMMING



MD Imthiaz I
192521360RD

CO2-AT1-Assignment1

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QUESTIONS

Q1

Smart Document Editor using
String, StringBuilder and
StringBuffer

10 marks

**Q2**

Hospital Token Management
System using List Interface

10 marks

**Q3**

Airport Boarding Queue
Management using Queue
Interface

10 marks

**Q4**

Digital Certificate Verification
using Set Interface

10 marks

**Q5**

Smart Inventory Management
using Map Interface

10 marks



5/5 answered

Q1

Smart Document Editor using String, StringBuilder and StringBuffer

10
marks

Question 1 :

- 1.Read a paragraph from the user.
- 2.Count the number of characters, words, and sentences.
- 3.Replace every occurrence of a given word with another word.
- 4.Reverse each sentence individually.
- 5.Maintain editing history using StringBuffer.
- 6.Construct the final formatted document using StringBuilder.
- 7.Display the original document, editing history, and final document.

Requirement: Use String methods wherever appropriate. Use StringBuilder for document construction and StringBuffer for edit history.

Your Code

```
1 import java.util.Scanner;  
2 public class DocumentEditor {  
3     public static void main(String[] args)  
4     {  
5         Scanner sc = new Scanner(System.in);  
6         System.out.println("Enter a paragraph");  
7         String paragraph = sc.nextLine();  
8         int characters = paragraph.length();  
9         String words[] = paragraph.trim().split("\\s+");
```

```

9      int wordCount = words.length;
10     String sentences[] = paragraph.split(" ");
11     int sentenceCount = 0;
12     for (String s : sentences) {
13         if (!s.trim().isEmpty())
14             sentenceCount++;
15     }
16     System.out.println("\nCharacters : " + wordCount);
17     System.out.println("Words          : " + sentenceCount);
18     System.out.println("Sentences       : " + sentenceCount);
19     System.out.print("\nEnter word to replace: ");
20     String oldWord = sc.next();
21     System.out.print("Enter new word: ");
22     String newWord = sc.next();
23     String replacedParagraph = paragraph.replace(oldWord, newWord);
24     StringBuffer history = new StringBuffer();
25     history.append("Original Paragraph\n");
26     history.append(paragraph);
27     history.append("\n\nReplaced '");
28     history.append(oldWord);
29     history.append("' with '");
30     history.append(newWord);
31     history.append("'\n");
32     history.append(replacedParagraph);
33     StringBuilder finalDocument = new StringBuilder();
34     String reverseSentences[] = replacedParagraph.split(" ");
35
36     for (String sentence : reverseSentences) {
37         String temp = sentence.trim();
38         if (!temp.isEmpty()) {
39             StringBuilder rev = new StringBuilder();
40             finalDocument.append(rev.append(temp).append(" "));
41         }
42     }
43     System.out.println("\n===== Original Paragraph =====");
44     System.out.println(paragraph);
45     System.out.println("\n===== Replaced Paragraph =====");
46     System.out.println(history);
47     System.out.println("\n===== Final Document =====");
48     System.out.println(finalDocument.toString());
49     sc.close();
50 }
51 }

```

Input

Java is easy to learn. Java is object oriented.
It is widely used!

Java
Python



Output

Enter a paragraph:

Characters : 66

Words : 13

Sentences : 3



Run



Save

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QUESTIONS

Q1

Smart Document Editor using String, StringBuilder and StringBuffer
10 marks

**Q2**

Hospital Token Management System using List Interface
10 marks

**Q3**

Airport Boarding Queue Management using Queue Interface
10 marks

**Q4**

Digital Certificate Verification using Set Interface
10 marks

**Q5**

Smart Inventory Management using Map Interface
10 marks



5/5 answered

Q2

Hospital Token Management System using List Interface

10
marks

Hospital Token Management System using List Interface

Store patient details using an appropriate List implementation.

Register new patients.

Display patients in registration order.

Update patient information using Patient ID.

Remove cancelled appointments.

Display all patients of a particular department.

Find the first and last registered patients.

Requirement: Use a suitable List implementation

Your Code

```
1 import java.util.ArrayList;
2 import java.util.List;
3 import java.util.Scanner;
4 public class HospitalTokenManagement {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8         List<Patient> patients = new ArrayList<>();
9         System.out.print("Enter number of patients: ");
10        int n = sc.nextInt();
```

```

10         sc.nextLine();
11         for (int i = 0; i < n; i++) {
12             System.out.println("\nEnter Patient Details:");
13             System.out.print("Patient ID: ");
14             int id = sc.nextInt();
15             sc.nextLine();
16             System.out.print("Name: ");
17             String name = sc.nextLine();
18             System.out.print("Department: ");
19             String dept = sc.nextLine();
20             patients.add(new Patient(id, name, dept));
21         }
22         System.out.println("\n--- Patients List ---");
23         for (Patient p : patients) {
24             p.display();
25         }
26         System.out.print("\nEnter Patient ID to Update: ");
27         int updateId = sc.nextInt();
28         sc.nextLine();
29         boolean found = false;
30         for (Patient p : patients) {
31             if (p.id == updateId) {
32                 System.out.print("Enter New Name: ");
33                 p.name = sc.nextLine();
34                 System.out.print("Enter New Department: ");
35                 p.department = sc.nextLine();
36                 found = true;
37                 System.out.println("Patient Updated Successfully");
38                 break;
39             }
40         }
41         if (!found) {
42             System.out.println("Patient ID not found");
43         }
44         System.out.print("\nEnter Patient ID to Remove: ");
45         int removeId = sc.nextInt();
46         found = false;
47         for (int i = 0; i < patients.size(); i++) {
48             if (patients.get(i).id == removeId) {
49                 patients.remove(i);
50                 found = true;
51                 System.out.println("Patient Removed Successfully");
52                 break;
53             }
54         }
55         if (!found) {
56             System.out.println("Patient ID not found");
57         }

```

```

58         sc.nextLine();
59         System.out.print("\nEnter Department: ");
60         String searchDept = sc.nextLine();
61         System.out.println("\nPatients in Department: ");
62         found = false;
63         for (Patient p : patients) {
64             if (p.department.equalsIgnoreCase(searchDept)) {
65                 p.display();
66                 found = true;
67             }
68         }
69         if (!found) {
70             System.out.println("No patient found in the department.");
71         }
72         if (!patients.isEmpty()) {
73             System.out.println("--- First Patient ---");
74             patients.get(0).display();
75             System.out.println("--- Last Patient ---");
76             patients.get(patients.size() - 1).display();
77         }
78         sc.close();
79     }
80 }
81 class Patient {
82     int id;
83     String name;
84     String department;
85
86     Patient(int id, String name, String department) {
87         this.id = id;
88         this.name = name;
89         this.department = department;
90     }
91     void display() {
92         System.out.println("Patient ID : " + id);
93         System.out.println("Name : " + name);
94         System.out.println("Department : " + department);
95         System.out.println();
96     }
97 }

```



Input

```

1
101
Arun
Cardiology

```



Output

```
Enter number of patients:
Enter Patient 1 Details
Patient ID: Name: Department:
--- Patients in Registration Order ---
Patient ID : 101
Name       : Arun
```

 Run Save

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JAVA PROGRAMMING

MD Imthiaz I
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Management using Queue
Interface

10 marks

**Q4**Digital Certificate Verification
using Set Interface

10 marks

**Q5**Smart Inventory Management
using Map Interface

10 marks



5/5 answered

Board passengers in FIFO order.

Display the next passenger without removing them.

Handle VIP passengers using a priority-based queue.

Display remaining passengers after each boarding operation.

Requirement: Use appropriate Queue implementations.

Your Code

```
1 import java.util.LinkedList;
2 import java.util.PriorityQueue;
3 import java.util.Queue;
4 import java.util.Scanner;
5 public class AirportBoardingQueue {
6     public static void main(String[] args)
7     {
8         Scanner sc = new Scanner(System.in);
9         Queue<String> boardingQueue = new
10         PriorityQueue<String>();
11         vipQueue = new
12         PriorityQueue<String>();
13         System.out.print("Enter number of
```



```

11     int n = sc.nextInt();
12     sc.nextLine();
13     for (int i = 0; i < n; i++) {
14         System.out.print("Enter Passenger Name: ");
15         boardingQueue.offer(sc.nextLine());
16     }
17     System.out.print("\nEnter number of VIP Passengers: ");
18     int m = sc.nextInt();
19     sc.nextLine();
20     for (int i = 0; i < m; i++) {
21         System.out.print("Enter VIP Passenger Name: ");
22         vipQueue.offer(sc.nextLine());
23     }
24     System.out.println("\n--- Normal Passengers Queue ---");
25     System.out.println(boardingQueue);
26     System.out.println("\n--- VIP Passengers Queue ---");
27     System.out.println(vipQueue);
28     if (!vipQueue.isEmpty()) {
29         System.out.println("\nNext VIP Passenger to Board: ");
30     }
31     if (!boardingQueue.isEmpty()) {
32         System.out.println("Next Normal Passenger to Board: ");
33     }
34     System.out.println("\n--- Boarding Process ---");
35     while (!vipQueue.isEmpty() || !boardingQueue.isEmpty()) {
36         if (!vipQueue.isEmpty()) {
37             System.out.println("Boarding VIP Passenger: ");
38         } else {
39             System.out.println("Boarding Normal Passenger: ");
40         }
41         System.out.println("Remaining VIP Passengers: ");
42         System.out.println("Remaining Normal Passengers: ");
43         System.out.println();
44     }
45     System.out.println("All passengers boarded successfully.");
46     sc.close();
47 }
48 }

```

Input

3
Arun
Priya
Rahul

Output

```
[Arun, Priya, Rahul]
```

```
--- VIP Queue ---
```

```
[Aman, Zoya]
```

```
Next VIP Passenger: Aman
```

**Run****Save**

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QUESTIONS

Q1

Smart Document Editor using
String, StringBuilder and
StringBuffer
10 marks

Q4

Digital Certificate Verification using Set Interface

10
marks

Digital Certificate Verification using Set Interface
Store certificate IDs using an appropriate Set
implementation



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JAVA PROGRAMMING

MD Imthiaz I
192521360RD**Q4**

Digital Certificate Verification
using Set Interface
10 marks

Q5

Smart Inventory Management
using Map Interface
10 marks

5/5 answered

Requirement: Use suitable Set implementations.

Your Code

```
1 import java.util.*;  
2 public class DigitalCertificateVerificatio  
3     public static void main(String[] args)  
4         Scanner sc = new Scanner(System.in)  
5         LinkedHashSet<String> certificates  
6         System.out.print("Enter the number  
7         int n = sc.nextInt();  
8         System.out.println("Enter Certific  
9         for (int i = 0; i < n; i++) {  
10             certificates.add(sc.next());  
11         }
```

```

12         System.out.println("\nCertificates
13         for (String cert : certificates) {
14             System.out.print(cert + " ");
15         }
16         System.out.print("\n\nEnter Certificate IDs:
17         String search = sc.next();
18         if (certificates.contains(search))
19             System.out.println("Certificate found");
20         } else {
21             System.out.println("Certificate not found");
22         }
23         TreeSet<String> sorted = new TreeSet<>();
24         System.out.println("\nCertificates in Sorted Order:");
25         for (String cert : sorted) {
26             System.out.print(cert + " ");
27         }
28         System.out.println("\n\nTotal Unique Certificates: " + certificates.size());
29         System.out.print("\n\nEnter Revoked Certificate ID: ");
30         String revoked = sc.next();
31         if (certificates.remove(revoked))
32             System.out.println("Certificate removed successfully");
33         } else {
34             System.out.println("Certificate not found");
35         }
36         System.out.println("\n\nRemaining Certificates:");
37         for (String cert : certificates) {
38             System.out.print(cert + " ");
39         }
40         sc.close();
41     }
42 }

```

Input

CERT104
CERT102
CERT103
CERT102

Output

Enter the number of certificate IDs: Enter
Certificate IDs:

Certificates in Insertion Order:
CERT101 CERT102 CERT103 CERT104

 Run

 Save

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JAVA PROGRAMMING



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QUESTIONS

Q1

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Q2

Hospital Token Management
System using List Interface

10 marks

Q3

Airport Boarding Queue
Management using Queue
Interface

10 marks

Q4

Digital Certificate Verification
using Set Interface

10 marks

Q5

Smart Inventory Management
using Map Interface

10 marks

5/5 answered

Q5

Smart Inventory Management using Map Interface

10

marks

Smart Inventory Management using Map Interface

Store product information using an appropriate Map
implementation.

Add new products.

Update product quantity.

Search a product using Product ID.

Calculate the total inventory value.

Display products category-wise.

Remove discontinued products.

Display all available products.

Requirement: Use a suitable Map implementation.

Your Code

```
1 import java.util.*;
2 public class SmartInventoryManagement {
3     public static void main(String[] args)
4     {
5         Scanner sc = new Scanner(System.in);
6         HashMap<Integer, Product> inventory = new HashMap<>();
7         System.out.print("Enter number of products: ");
8         int n = sc.nextInt();
9         for (int i = 0; i < n; i++) {
10             System.out.println("\nEnter Product ID and Name: ");
```

```

10         int id = sc.nextInt();
11         sc.nextLine();
12         System.out.print("Enter Product Name: ");
13         String name = sc.nextLine();
14         System.out.print("Enter Category: ");
15         String category = sc.nextLine();
16         System.out.print("Enter Quantity: ");
17         int quantity = sc.nextInt();
18         System.out.print("Enter Price: ");
19         double price = sc.nextDouble();
20         inventory.put(id, new Product(id, name, category, quantity, price));
21     }
22     System.out.print("\nEnter Product ID to Update: ");
23     int updateId = sc.nextInt();
24     if (inventory.containsKey(updateId)) {
25         System.out.print("Enter New Quantity: ");
26         inventory.get(updateId).quantity = sc.nextInt();
27         System.out.println("Quantity Updated Successfully");
28     } else {
29         System.out.println("Product ID Not Found");
30     }
31     System.out.print("\nEnter Product ID to Search: ");
32     int searchId = sc.nextInt();
33     if (inventory.containsKey(searchId)) {
34         Product p = inventory.get(searchId);
35         System.out.println("Product Found");
36         System.out.println(p.name + " | " + p.category + " | " + p.quantity + " | " + p.price);
37     } else {
38         System.out.println("Product ID Not Found");
39     }
40     double total = 0;
41     for (Product p : inventory.values()) {
42         total += p.quantity * p.price;
43     }
44     System.out.println("\nTotal Inventory Value: " + total);
45     System.out.println("\nProducts by Category");
46     LinkedHashMap<String, ArrayList<Product>> categoryMap = new LinkedHashMap<>();
47     for (Map.Entry<Integer, Product> entry : inventory.entrySet()) {
48         Product p = entry.getValue();
49         if (!categoryMap.containsKey(p.category)) {
50             categoryMap.put(p.category, new ArrayList<>());
51         }
52         categoryMap.get(p.category).add(p);
53     }
54     for (String cat : categoryMap.keySet()) {
55         System.out.println(cat + " : ");
56     }
57     System.out.print("\nEnter Product ID to Delete: ");

```

```

58         int removeId = sc.nextInt();
59         if (inventory.remove(removeId) !=
60             System.out.println("Product Re
61     } else {
62         System.out.println("Product No
63     }
64     System.out.println("\nAvailable Pr
65     for (Map.Entry<Integer, Product> e
66         Product p = entry.getValue();
67         System.out.println(entry.getKe
68             " Qty:" + p.quantity -
69     }
70     sc.close();
71 }
72 }
73 class Product {
74     String name;
75     String category;
76     int quantity;
77     double price;
78     Product(String name, String category,
79         this.name = name;
80         this.category = category;
81         this.quantity = quantity;
82         this.price = price;
83     }
84 }
85

```

Input

2000
102
15
101
103

Output

Enter number of products:
Enter Product ID:
Enter Product Name: Enter Category: Enter
Quantity: Enter Price:
Enter Product ID:
Enter Product Name: Enter Category: Enter

▶ Run

⬆ Save

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